

Amna Ahmad

Researcher / Developer

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PROFESSIONAL EXPERIENCE

RESEARCHER

Des Moines, Iowa

World Food Prize

2024-Present

- Developed a fully offline AI-powered application to assist community health workers in diagnosing and treating SAM in resource-limited settings.
- System generates real-time treatment plans based on patient input (e.g., age, weight, MUAC), with multilingual support and optional data upload.
- Selected as a **Global Challenge Grand Prize Winner** for the most innovative solution in combating global hunger.

Software Engineering Intern

CHAI CORE- UNC-CHAPEL-HILL

- Collaborate with a cross-functional team of engineers and researchers to develop custom software solutions for UNC faculty and grantees.
- Translate research workflows into scalable digital tools that improve data collection, analysis, and reporting for public health and social science projects.
- Participate in full software lifecycle—from requirements gathering to design, development, testing, and deployment—ensuring accessibility and user-friendliness.
- Contribute to both frontend and backend development, utilizing tools such as JavaScript, Python, and cloud-based services to support impactful grant-funded research.

PERSONAL PROJECTS

RTPS- Real Time Patent Search

July 2025

- Developed a semantic search system using Retrieval-Augmented Generation (RAG) to answer complex patent-related questions by retrieving and citing relevant text chunks from a vector database.
- Designed custom embeddings and PostgreSQL-backed search pipelines to support accurate context-aware retrieval from a patent corpus.
- Integrated Redis for caching and latency reduction, and wrapped the retrieval results into GPT-based completions with structured citations.
- Built the system to simulate a patent expert's workflow, enabling natural language Q&A over technical documents
- I built this because of the difficulty I was having understanding existing prior-art and patents when creating my malnutrition treatment app for the World Food Prize.

EDUCATION

GREEN HOPE HIGH SCHOOL

Cary, North Carolina

Graduated Summa Cum Laude

2021-2025